

Thermodynamics Formula Sheet

Chapter 2

Pressure

Constants

$$\begin{aligned}1 \text{ atm} &= 101.325 \text{ kPa} \\&= 1.01325 \text{ bar} \\&= 760 \text{ Torr} \\&= 760 \text{ mmHg} \\&= 29.92 \text{ inHg} \\&= 14.696 \text{ psi} \\&= 10.33 \text{ m H}_2\text{O (at 4}^\circ\text{C)}\end{aligned}$$

Pressure Conversions

$$\begin{aligned}1 \text{ Pa} &= 1 \text{ N/m}^2 \\1 \text{ bar} &= 10^5 \text{ Pa} \\1 \text{ psi} &= 6.89476 \times 10^3 \text{ Pa} \\1 \text{ Torr} &= 133.322 \text{ Pa} \\1 \text{ mmHg} &= 133.322 \text{ Pa} \\1 \text{ inHg} &= 3.38639 \times 10^3 \text{ Pa}\end{aligned}$$

Gauge vs Absolute Pressure

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmospheric}}$$

$$P_{\text{vacuum}} = P_{\text{atmospheric}} - P_{\text{absolute}}$$

Note: Generally, gauge pressures already account for atmospheric pressure, and thus reads zero when open to the atmosphere.

Pressure Formulas

$$P = F/A$$

$$P = \rho gh \quad (\text{hydrostatic pressure})$$

Temperature

Temperature Conversions

- Absolute temperature conversions
 - $T_K = T_{^\circ C} + 273.15$
 - $T_{^\circ R} = T_{^\circ F} + 459.67$
 - $T_{^\circ F} = \frac{9}{5}T_{^\circ C} + 32$
 - $T_{^\circ R} = \frac{9}{5}T_K$
- Temperature difference conversions
 - $\Delta K = \Delta ^\circ C$
 - $\Delta ^\circ R = \Delta ^\circ F$
 - $\Delta ^\circ F = \frac{9}{5}\Delta ^\circ C$
 - $\Delta ^\circ R = \frac{9}{5}\Delta K$

SI Prefixes

$$\begin{aligned}\text{femto (f)} &= 10^{-15} \\ \text{pico (p)} &= 10^{-12} \\ \text{nano (n)} &= 10^{-9} \\ \text{micro } (\mu) &= 10^{-6} \\ \text{milli (m)} &= 10^{-3} \\ \text{centi (c)} &= 10^{-2} \\ \text{deci (d)} &= 10^{-1} \\ \text{deca (da)} &= 10^1 \\ \text{hecto (h)} &= 10^2 \\ \text{kilo (k)} &= 10^3 \\ \text{mega (M)} &= 10^6 \\ \text{giga (G)} &= 10^9 \\ \text{tera (T)} &= 10^{12} \\ \text{peta (P)} &= 10^{15}\end{aligned}$$

Note: 1Angstrom ($\text{\AA}$) = 10^{-10} m

Density and Specific Gravity

$$\rho = \frac{m}{V} \quad (\text{density})$$

$$\vartheta = \frac{V}{m} = \frac{1}{\rho} \quad (\text{specific volume})$$

Specific gravity is defined as a relative density compared to water (at 4°C) where $\rho_{\text{water}} = 1000 \text{ kg/m}^3$:

$$SG = \frac{\rho_{\text{fluid}}}{\rho_{\text{water}}} \implies \rho_{\text{fluid}} = SG \cdot \rho_{\text{water}}$$

Specific Weight

$$\gamma_s = \rho g \quad (\text{specific weight})$$

Where hydrostatic pressure can be written as $P = \gamma_s h$

Slugs and lbf

$$\left[\frac{1 \text{ lbf}}{32.174 \frac{\text{lbm} \cdot \text{ft}}{\text{s}^2}} \right], \quad \left[\frac{1 \text{ slug}}{32.174 \text{ lb}} \right], \quad \left[\frac{1 \text{ lbf}}{1 \frac{\text{slug} \cdot \text{ft}}{\text{s}^2}} \right]$$

Energy Units

$$1 \text{ Btu} = 778.17 \text{ ft} \cdot \text{lbf} = 1055.06 \text{ J}$$

Work Units

$$1 \text{ hp} = 550 \text{ ft} \cdot \text{lbf/s} = 745.7 \text{ W}$$

Power

Power ($\dot{W}$) is the rate of work done, and can be calculated for a control volume (open system with mass flow) using the first law of thermodynamics:

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i \right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e \right]$$

Where:

- dE/dt is the rate of change of the total energy of the system over time
- $\dot{Q}$ is the rate of heat transfer into the system from the surroundings
- $\dot{W}$ is the rate of work done by the system on its surroundings (power output)
- The summations represent the total energy entering and leaving the system due to mass flow, where h is the specific enthalpy, $v/2$ is the specific kinetic energy, and gz is the specific potential energy at the inlet (i) and outlet (e) of the control volume.

For a steady state process ($dE/dt = 0$) with no heat transfer ($\dot{Q} = 0$), the equation becomes

$$\dot{W} = \dot{m}e_{\text{mech}}$$
$$e_{\text{mech}} = \frac{P}{\rho} + \frac{v^2}{2} + gz \quad (\text{P is pressure})$$

Where e_{mech} is the mechanical energy per unit mass, and $\dot{m}$ is the mass flow rate.

$$\dot{m} = \rho v A$$

$$\dot{m} = \rho \dot{V}$$

Note: $\dot{V}$ is the volumetric flow rate, and A is the cross-sectional area of the flow.
From physics, power is also defined as

$$\dot{W} = F \cdot v$$

Where F is the force applied to an object and v is the velocity of the object.

Fluid Flow

$$\dot{W} = \dot{V}(P_e - P_i)$$

Spring

$$\dot{W}_{\text{spring}} = \frac{1}{2}k(x_f^2 - x_i^2)$$

Heat Transfer by Conduction

$$\dot{Q}_{\text{cond}} = kA \frac{\Delta T}{\Delta x}$$

Where: $\Delta T = T_i - T_e$, A is the area of the window, k is the thermal conductivity of the material, and Δx is the thickness of the material. The amount of heat transferred is then given by

$$Q_{\text{cond}} = \dot{Q}_{\text{cond}} \cdot t$$

Where t is the time duration of heat transfer.